

EE133 Solid-State Electronics

Homework 7

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Problem 1 (50 pts) — Finding the Bandgap

Setup

The diode current density under forward bias is:

$$J \approx J_0 e^{qV_A/kT}$$

where J_0 depends on the minority carrier concentrations, which are set by n_i^2 . Since $n_i^2 \propto T^3 e^{-E_g/kT}$, and diffusion coefficients D and lifetimes τ are given to be independent of T , the diffusion lengths $L = \sqrt{D\tau}$ are also T -independent. Therefore:

$$J_0(T) \propto n_i^2(T) \propto T^3 e^{-E_g/kT}$$

and the total current is:

$$J(T) \propto T^3 e^{-E_g/kT} \cdot e^{qV_A/kT} = T^3 e^{(qV_A - E_g)/kT}$$

Calculation

Taking the ratio at $T_H = 300$ K and $T_L = 250$ K with $V_A = 0.7$ V:

$$\frac{J(T_H)}{J(T_L)} = \left(\frac{T_H}{T_L}\right)^3 \exp\left[(qV_A - E_g) \left(\frac{1}{kT_H} - \frac{1}{kT_L}\right)\right] = 375$$

Computing each factor:

$$\left(\frac{300}{250}\right)^3 = (1.2)^3 = 1.728$$

$$\frac{1}{kT_H} - \frac{1}{kT_L} = \frac{1}{k} \left(\frac{1}{300} - \frac{1}{250}\right) = \frac{1}{k} \cdot \frac{250 - 300}{300 \times 250} = \frac{-1}{1500k}$$

with $k = 8.617 \times 10^{-5}$ eV/K, so $1500k = 0.12926$ eV.

Therefore:

$$\frac{375}{1.728} = e^{(qV_A - E_g)/0.12926} \implies \ln(217.0) = \frac{0.7 \text{ eV} - E_g}{0.12926 \text{ eV}}$$

$$5.380 = \frac{0.7 - E_g}{0.12926} \implies 0.7 - E_g = -0.695 \text{ eV}$$

$$\boxed{E_g \approx 1.40 \text{ eV}}$$

This is close to the bandgap of GaAs ($E_g = 1.42$ eV), which is physically consistent.

Problem 2 — Temperature Effects on Diode Current

(a) Forward Bias — Does current increase or decrease?

Under forward bias, the diode current is:

$$J \approx J_0 e^{qV_A/kT}$$

As derived in Problem 1, $J \propto T^3 e^{(qV_A - E_g)/kT}$. Since $E_g > qV_A$ (typically $E_g \approx 1.1$ eV for Si, $V_A \approx 0.7$ V), the exponent $(qV_A - E_g)/kT$ is *negative*, and as T increases, its magnitude $|qV_A - E_g|/kT$ *decreases*, making $e^{(qV_A - E_g)/kT}$ larger. The T^3 prefactor also increases.

Therefore, current increases with temperature under forward bias. Physically, increasing T generates more intrinsic carriers (n_i increases exponentially), which dramatically increases J_0 and hence the forward current.

(b) Reverse Bias — Does current increase or decrease?

Under reverse bias ($V_A \rightarrow -|V_A| \gg kT/q$):

$$J \approx -J_0 \propto -T^3 e^{-E_g/kT}$$

As T increases, $n_i^2 \propto T^3 e^{-E_g/kT}$ increases rapidly (the exponential dominates). Since $J_0 \propto n_i^2$, the magnitude of the reverse current increases.

Therefore, the reverse-bias current magnitude increases with temperature. The reverse saturation current doubles approximately every 10 K for silicon, because $J_0 \propto n_i^2 \propto e^{-E_g/kT}$ is extremely temperature-sensitive.

Problem 3 — Doubling N_D in a p^+ - n Junction

In a p^+ - n junction, $N_A \gg N_D$, so N_A dominates. The relevant formulas for this one-sided junction are:

$$V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}, \quad W \approx x_n \approx \sqrt{\frac{2\epsilon_s V_{bi}}{q N_D}}, \quad C_{j0} = \frac{\epsilon_s}{W} \propto \sqrt{N_D},$$

$$J_0 = \frac{q D_p n_i^2}{L_p N_D} \quad (p^+-n, \text{ minority holes in } n\text{-side dominate})$$

(a) Junction Capacitance — Increases

$$C_{j0} = \sqrt{\frac{q\epsilon_s N_D}{2V_{bi}}}$$

Doubling N_D increases C_{j0} . Although V_{bi} increases slightly (logarithmically), the $\sqrt{N_D}$ dependence dominates. **Junction capacitance increases.**

(b) Built-in Potential — Increases

$$V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$

Doubling N_D adds $\frac{kT}{q} \ln 2 \approx 18 \text{ mV}$ at 300 K. V_{bi} **increases (slightly)**.

(c) Breakdown Voltage — Increases

The avalanche breakdown voltage for a one-sided p^+-n junction is approximately:

$$V_{Br} \approx \frac{\varepsilon_s E_{crit}^2}{2qN_D}$$

where E_{crit} is roughly constant (weakly dependent on doping). Doubling N_D *decreases* the numerator-to-denominator ratio, so V_{Br} **decreases**.

(Note: More precisely, increasing N_D raises the peak electric field for a given reverse bias, triggering avalanche at a lower voltage.)

(d) Ohmic Losses — Decrease

Ohmic losses arise from the series resistance of the bulk semiconductor regions. The conductivity of the n -side is:

$$\sigma_n = q\mu_n N_D$$

Doubling N_D roughly doubles σ_n (mobility decreases slightly due to impurity scattering, but the increase in carrier density dominates). Lower resistance means **Ohmic losses decrease**.

(e) Diode Diffusion Current at Fixed Positive Voltage — Decreases

For a p^+-n junction, the saturation current is dominated by minority hole injection into the n -side:

$$J_0 = \frac{qD_p n_i^2}{L_p N_D}$$

Doubling N_D halves J_0 (approximately), which means the forward diffusion current $J = J_0(e^{qV_A/kT} - 1)$ **decreases**.

Problem 4 — Band Diagram Analysis

Given: $F_p - F_n = 0.5 \text{ eV}$, $\Delta E_c = 1.3 \text{ eV}$ (far left to far right), $T = 300 \text{ K}$, $n_i = 10^{10} \text{ cm}^{-3}$, $\varepsilon_r = 11.8$, $x_p = x_n = W/2$ (symmetric junction).

(a) Forward or Reverse Biased?

Reverse biased. In the band diagram, the quasi-Fermi level F_p lies *above* F_n throughout the junction region, meaning the splitting of quasi-Fermi levels has the sign associated with reverse bias (electron quasi-Fermi level depressed below hole quasi-Fermi level in the depletion region). Under reverse bias, the total band bending exceeds the equilibrium built-in potential.

(b) Magnitude of Applied Bias

The separation of the quasi-Fermi levels equals the applied voltage:

$$|V_A| = \frac{F_p - F_n}{q} = \frac{0.5 \text{ eV}}{q} \implies \boxed{|V_A| = V_R = 0.5 \text{ V}}$$

(c) Built-in Voltage

For a reverse-biased junction, the total band bending (Ec difference from far left to far right) is:

$$\begin{aligned} \Delta E_c &= q(V_{bi} + V_R) \\ 1.3 \text{ eV} &= q(V_{bi} + 0.5 \text{ V}) \implies V_{bi} = 1.3 - 0.5 = \boxed{0.8 \text{ V}} \end{aligned}$$

(d) Values of N_D and N_A

Since $x_p = x_n = W/2$, by the charge neutrality condition $N_A x_p = N_D x_n$, we must have $N_A = N_D$.

Using the built-in voltage for a symmetric junction:

$$\begin{aligned} V_{bi} &= \frac{2kT}{q} \ln \frac{N_A}{n_i} \\ 0.8 \text{ V} &= 2 \times 0.02585 \text{ V} \times \ln \frac{N_A}{10^{10}} \\ \ln \frac{N_A}{10^{10}} &= \frac{0.8}{0.05170} = 15.47 \\ N_A &= 10^{10} \cdot e^{15.47} = 10^{10} \times 5.24 \times 10^6 \end{aligned}$$

$$\boxed{N_A = N_D \approx 5.24 \times 10^{16} \text{ cm}^{-3}}$$

(e) Values of C_{j0} and C_j

First find the depletion width at zero bias W_0 and at the operating point W .

For a symmetric junction, the total depletion width is:

$$W = \sqrt{\frac{4\epsilon_s(V_{bi} + V_R)}{qN_D}}$$

Using SI units: $\varepsilon_s = 11.8 \times 8.854 \times 10^{-12} = 1.045 \times 10^{-10} \text{ F/m}$, $N_D = 5.24 \times 10^{22} \text{ m}^{-3}$, $q = 1.6 \times 10^{-19} \text{ C}$.

At zero bias ($V_{bi} = 0.8 \text{ V}$):

$$W_0 = \sqrt{\frac{4 \times 1.045 \times 10^{-10} \times 0.8}{1.6 \times 10^{-19} \times 5.24 \times 10^{22}}} = \sqrt{\frac{3.344 \times 10^{-10}}{8.384 \times 10^3}} = \sqrt{3.988 \times 10^{-14}} = 2.00 \times 10^{-7} \text{ m}$$

$$C_{j0} = \frac{\varepsilon_s}{W_0} = \frac{1.045 \times 10^{-10}}{2.00 \times 10^{-7}} = \boxed{5.23 \times 10^{-4} \text{ F/m}^2 \approx 52.3 \text{ nF/cm}^2}$$

At operating bias ($V_{bi} + V_R = 1.3 \text{ V}$):

$$C_j = C_{j0} \sqrt{\frac{V_{bi}}{V_{bi} + V_R}} = 5.23 \times 10^{-4} \times \sqrt{\frac{0.8}{1.3}} = 5.23 \times 10^{-4} \times 0.784 = \boxed{4.10 \times 10^{-4} \text{ F/m}^2}$$

Equivalently:

$$W = W_0 \sqrt{\frac{V_{bi} + V_R}{V_{bi}}} = 2.00 \times 10^{-7} \times \sqrt{\frac{1.3}{0.8}} = 2.55 \times 10^{-7} \text{ m}$$

$$C_j = \varepsilon_s / W = 1.045 \times 10^{-10} / 2.55 \times 10^{-7} = 4.10 \times 10^{-4} \text{ F/m}^2 \quad \checkmark$$

(f) Voltage to Obtain $C_{j0}/2$

We need:

$$C_j = \frac{C_{j0}}{\sqrt{1 + V_R/V_{bi}}} = \frac{C_{j0}}{2}$$

(where $V_R > 0$ for reverse bias, and for forward bias we replace V_R with $-V_A$).

$$\sqrt{1 + V_R/V_{bi}} = 2 \implies 1 + V_R/V_{bi} = 4 \implies V_R = 3V_{bi} = 3 \times 0.8 \text{ V} = 2.4 \text{ V}$$

The sign is **negative** (reverse bias), and the magnitude is:

$$\boxed{V_A = -2.4 \text{ V} \quad (\text{i.e., } V_R = 2.4 \text{ V reverse bias})}$$

(g) Minority Carrier Distributions at Depletion Edges

The equilibrium minority carrier concentrations are:

$$p_{n0} = \frac{n_i^2}{N_D} = \frac{(10^{10})^2}{5.24 \times 10^{16}} = \frac{10^{20}}{5.24 \times 10^{16}} \approx 1.91 \times 10^3 \text{ cm}^{-3}$$

$$n_{p0} = \frac{n_i^2}{N_A} = p_{n0} \approx 1.91 \times 10^3 \text{ cm}^{-3} \quad (N_A = N_D)$$

Under reverse bias $V_R = 0.5 \text{ V}$, the minority carrier concentrations at the depletion edges are:

$$p_n(+W/2) = p_{n0} e^{-qV_R/kT} = 1.91 \times 10^3 \times e^{-0.5/0.02585} = 1.91 \times 10^3 \times e^{-19.34} \approx 7.6 \times 10^{-6} \text{ cm}^{-3} \approx 0$$

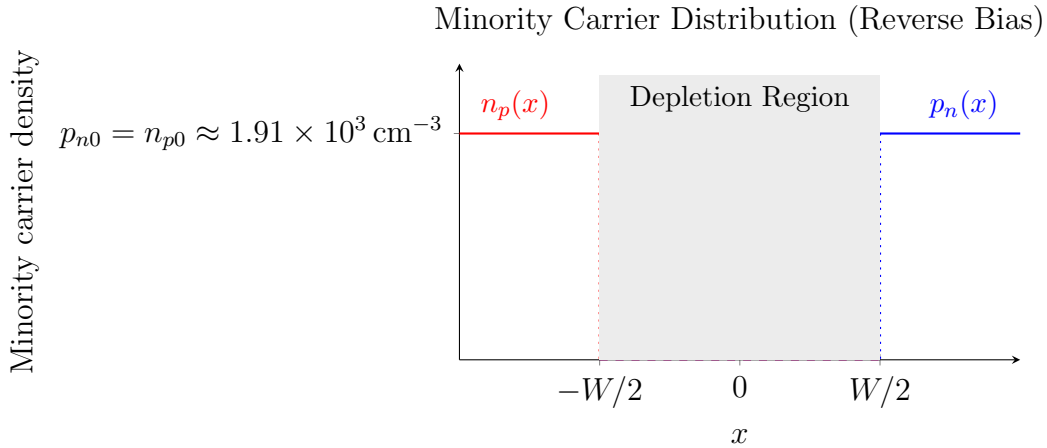
$$n_p(-W/2) \approx 0 \quad (\text{same calculation by symmetry})$$

Therefore, the excess minority carrier concentrations at the depletion edges are:

$$\Delta p_n = p_n(+W/2) - p_{n0} \approx -p_{n0} \approx -1.91 \times 10^3 \text{ cm}^{-3}$$

$$\Delta n_p = n_p(-W/2) - n_{p0} \approx -n_{p0} \approx -1.91 \times 10^3 \text{ cm}^{-3}$$

The minority carrier concentrations are essentially zero at the depletion edges (deeply reverse biased), decreasing from their bulk equilibrium values as we approach the junction from either side.



Under reverse bias the minority carriers at the edges of the depletion region are pulled essentially to zero; minority carriers are “extracted.”

(h) Current Density

Given: $D_n = 26 \text{ cm}^2/\text{s}$, $D_p = 13 \text{ cm}^2/\text{s}$, $\tau_n = \tau_p = 1 \mu\text{s}$.

Diffusion lengths:

$$L_p = \sqrt{D_p \tau_p} = \sqrt{13 \text{ cm}^2/\text{s} \times 10^{-6} \text{ s}} = \sqrt{1.3 \times 10^{-5}} = 3.606 \times 10^{-3} \text{ cm}$$

$$L_n = \sqrt{D_n \tau_n} = \sqrt{26 \times 10^{-6}} = 5.099 \times 10^{-3} \text{ cm}$$

The saturation current density for this symmetric junction is:

$$J_0 = qn_i^2 \left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} \right)$$

Computing each term with $N_A = N_D = 5.24 \times 10^{16} \text{ cm}^{-3}$:

$$\frac{D_p}{L_p N_D} = \frac{13}{3.606 \times 10^{-3} \times 5.24 \times 10^{16}} = \frac{13}{1.890 \times 10^{14}} = 6.88 \times 10^{-14} \text{ cm}^4/\text{s}$$

$$\frac{D_n}{L_n N_A} = \frac{26}{5.099 \times 10^{-3} \times 5.24 \times 10^{16}} = \frac{26}{2.672 \times 10^{14}} = 9.73 \times 10^{-14} \text{ cm}^4/\text{s}$$

$$\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A} = 1.661 \times 10^{-13} \text{ cm}^4/\text{s}$$

$$J_0 = (1.6 \times 10^{-19})(10^{10})^2(1.661 \times 10^{-13}) = (1.6 \times 10^{-19})(10^{20})(1.661 \times 10^{-13})$$

$$J_0 = 1.6 \times 1.661 \times 10^{-13} \times 10 = 2.66 \times 10^{-12} \text{ A/cm}^2$$

The total current density under reverse bias $V_R = 0.5 \text{ V}$ is:

$$J = J_0(e^{qV_A/kT} - 1) = J_0(e^{-19.34} - 1) \approx -J_0$$

$$J \approx -J_0 \approx -2.66 \times 10^{-12} \text{ A/cm}^2$$

The current flows in the reverse direction (out of the p -side), consistent with reverse-bias operation.

Problem 5 — Breakdown in Ge vs. GaAs

(a) Avalanche Breakdown

Avalanche breakdown occurs when the electric field is high enough to allow carriers to gain sufficient energy between collisions to ionize lattice atoms. The critical field $\mathcal{E}_{crit}$ is larger for wider-bandgap materials (carriers need more energy to ionize), and the breakdown voltage for a one-sided junction scales as $V_{Br} \propto \mathcal{E}_{crit}^2/N$.

Since $E_g(\text{GaAs}) = 1.42 \text{ eV} \gg E_g(\text{Ge}) = 0.66 \text{ eV}$, GaAs has a much larger critical field. Therefore, **GaAs has a larger avalanche breakdown voltage magnitude.**

(b) Zener (Band-to-Band Tunneling) Breakdown

Zener breakdown involves quantum mechanical tunneling of electrons across the bandgap from the valence band to the conduction band. The tunneling probability decreases exponentially with increasing bandgap. A larger bandgap makes tunneling much less probable, requiring a larger electric field (and hence larger reverse voltage) to achieve significant tunnel current.

Therefore, **GaAs has a larger Zener breakdown voltage magnitude** because its wider bandgap suppresses band-to-band tunneling.

Problem 6 — Decreasing Breakdown Voltage

Goal: Reduce V_{Br} from 8 V to 6 V in a p^+-n diode.

For a p^+-n one-sided junction:

$$V_{Br} \approx \frac{\varepsilon_s \mathcal{E}_{crit}^2}{2qN_D}$$

To **decrease** V_{Br} , we need to **increase** N_D (the n -side doping). A higher N_D compresses the depletion region, increasing the peak electric field at a given reverse voltage, so avalanche (or Zener) breakdown is triggered at a lower applied voltage.

Action: Increase the n -side doping N_D .

Quantitatively, since $V_{Br} \propto N_D^{-1}$ (ignoring the weak doping dependence of $\mathcal{E}_{crit}$):

$$\frac{V_{Br,new}}{V_{Br,old}} = \frac{N_{D,old}}{N_{D,new}} \implies N_{D,new} = N_{D,old} \times \frac{8}{6} = \frac{4}{3} N_{D,old}$$

So increasing N_D by approximately 33% would reduce V_{Br} from 8 V to 6 V.

Problem 7 — Current Mechanisms from I-V Curve

The $\log(I)$ vs. V plot shows four distinct regions with ideality factors $n = 2$, $n = 1$, $n = 2$, and saturation (region D).

(a) Region A ($n = 2$, low forward bias)

Dominant mechanism: Shockley–Read–Hall (SRH) generation–recombination current in the depletion region.

At low forward bias, the current is dominated by carrier recombination via mid-gap trap states in the depletion region:

$$J_{rec} \approx \frac{qn_i W}{2\tau_0} e^{qV_A/2kT}$$

This gives an ideality factor $n = 2$.

(b) Region B ($n = 1$, moderate forward bias)

Dominant mechanism: Ideal diffusion (injection) current.

At moderate forward bias, the recombination current is overtaken by the ideal diode diffusion current:

$$J_{diff} = J_0(e^{qV_A/kT} - 1)$$

This gives the ideal ideality factor $n = 1$, as minority carriers injected across the junction recombine in the quasi-neutral regions.

(c) Region C ($n = 2$, high forward bias)

Dominant mechanism: High-level injection (or series resistance effects).

At large forward bias, the injected minority carrier concentration becomes comparable to the majority carrier concentration (high-level injection). In this regime, both carrier types must increase together to maintain charge neutrality, leading to:

$$J \propto e^{qV_A/2kT}$$

giving $n = 2$. Alternatively, resistive voltage drop across the bulk semiconductor reduces the effective junction voltage, also causing $n \rightarrow 2$.

(d) Region D — Why the curve flattens

In region D (very high forward bias), the current no longer grows exponentially with voltage. This is because the **series resistance** of the diode (contact resistance + bulk semiconductor resistance) becomes significant. The voltage drop across this resistance IR_s grows linearly with current, so the junction voltage saturates, and the I-V curve transitions from exponential to linear (ohmic) behavior in the log plot, appearing flat.

(e) More Likely Semiconductor: Si or Ge?

Silicon (Si) is more likely.

The I-V curve shows a prominent region A ($n = 2$ SRH recombination) that is *wider* relative to region B. This means the recombination current dominates over the diffusion current over a large voltage range. The diffusion current $J_0 \propto n_i^2$ is much smaller for Si ($E_g = 1.12$ eV, $n_i \approx 10^{10} \text{ cm}^{-3}$) than for Ge ($E_g = 0.66$ eV, $n_i \approx 10^{13} \text{ cm}^{-3}$).

In Ge, the much larger n_i makes J_0 large enough that diffusion current dominates even at low voltages, suppressing the $n = 2$ region. In Si, J_0 is small enough that the recombination current (region A) is prominent before the diffusion current takes over (region B). **Therefore, this diode is most likely made from Si.**